UNIVERSITY OF UYO
FACULTY OF SCIENCE
DEPARTMENT OF MATHEMATICS

FIRST SEMESTER EXAMINATION
SESSION: 2022/2023
COURSE CODE: MTH311(MATHEMATICAL METHODS III)
TIME: 3HRS
INSTRUCTION: ANSWER ANY FOUR (4) QUESTIONS ONLY

1a. State the conditions that guarantee the existence and uniqueness of solution of the initial value problem (1.0) on the interval $a \le x \le b$

$$a_0(x)y'' + a_1(x)y' + a_2(x)y = f(x)$$
$$y(x_0) = y_0,\ y''(x_0) = y_0$$

bi. Using the Picard's iteration method approximate the solution of the initial value problem

$$y'(x) = 2x(1-y),\ y(0) = 0$$

bii. Show that the solution exists and is unique.

2a. Evaluate the eigenvalues and the corresponding eigen functions of the Sturm-Liouville problem

$$y'' + \lambda y = 0,\ y(0) = y(l) = 0 \qquad 2.0$$

b. Show that the eigen functions of equation (2.0) are orthogonal

3. Evaluate the series solution of the initial value problem $y'' + y = 0,\ y(0) = 0,\ y'(0) = 1$ near the point $x_0 = 0$

4a. Show that the following functions satisfies the corresponding partial differential equations

i $u = f(x^2 - y^2),\quad yu_x + xu_y = 0$ ii. $u = (x+a)(y+b),\ u_x u_y = u$

b. Solve the Euler equation $x^2y'' + 5xy' + y = 0$ near the $x_0 = 0$

5a. Evaluate the series solution of the differential equation near $xy'' - y = 0$ near the point $x_0 = 0$.

b. Show that the solution exists and is unique

6a. Evaluate the general solution of the Bessel equation $x^2y'' + xy' + (x^2 - 4)y = 0$ about the analytic point $x_0 = 0$.

6b. Compute the first three terms of the general solution

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2022/2023 FIRST SEMESTER TEST: MTH 316: INTRODUCTION TO MATHEMATICAL MODELLING
Instruction: Answer **ALL**
Time: 30mins

1 Describe the Causal-Relationships that exist between any three variables obtainable in a closed system of real world advertising problem using matrix tabular display.

2 Substance A is converted into substance B at a rate proportional to the square of the amount of substance A present at any time t. If 50 grams of Substance A was present at the beginning and only 10 grams was left unconverted after 2 hours. How much of substance A will remain after an additional 2 hours?

3 Let $u_n = su_{n-1} + t$ be a given recurrence relation, obtain its general solution.